

# California Housing Price Predictor

Linear Regression Model

# Objective:

Build and evaluate a Linear Regression model to predict median house prices in California

# Dataset:

1.Dataset Information

2.Source:California Housing Dataset(scikit-learn)

3.samples:20,640

4.Features:8

5.Target:Median House Value(MedHouseVal)

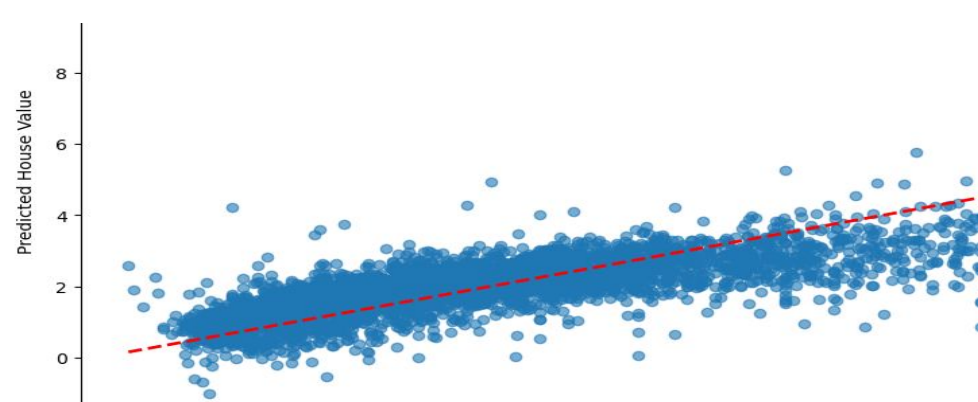
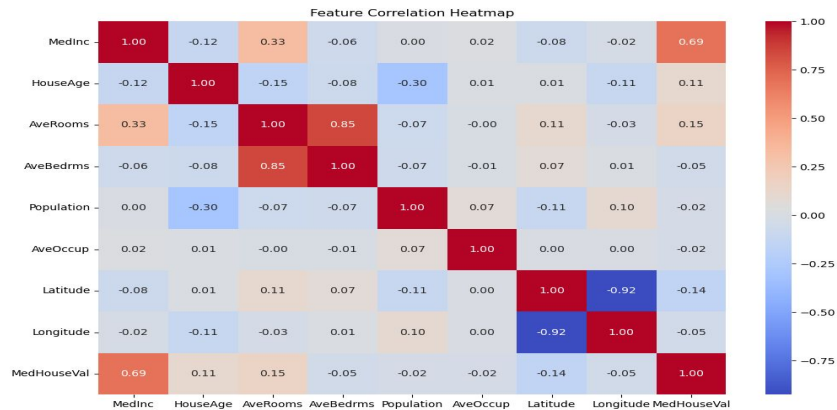
# EDA Highlights

Exploratory Data Analysis

Correlation Heatmap

Median Income vs House Value scatter plot

Key Insight: MedInc has strongest correlation



# Model Performance

## Model Results

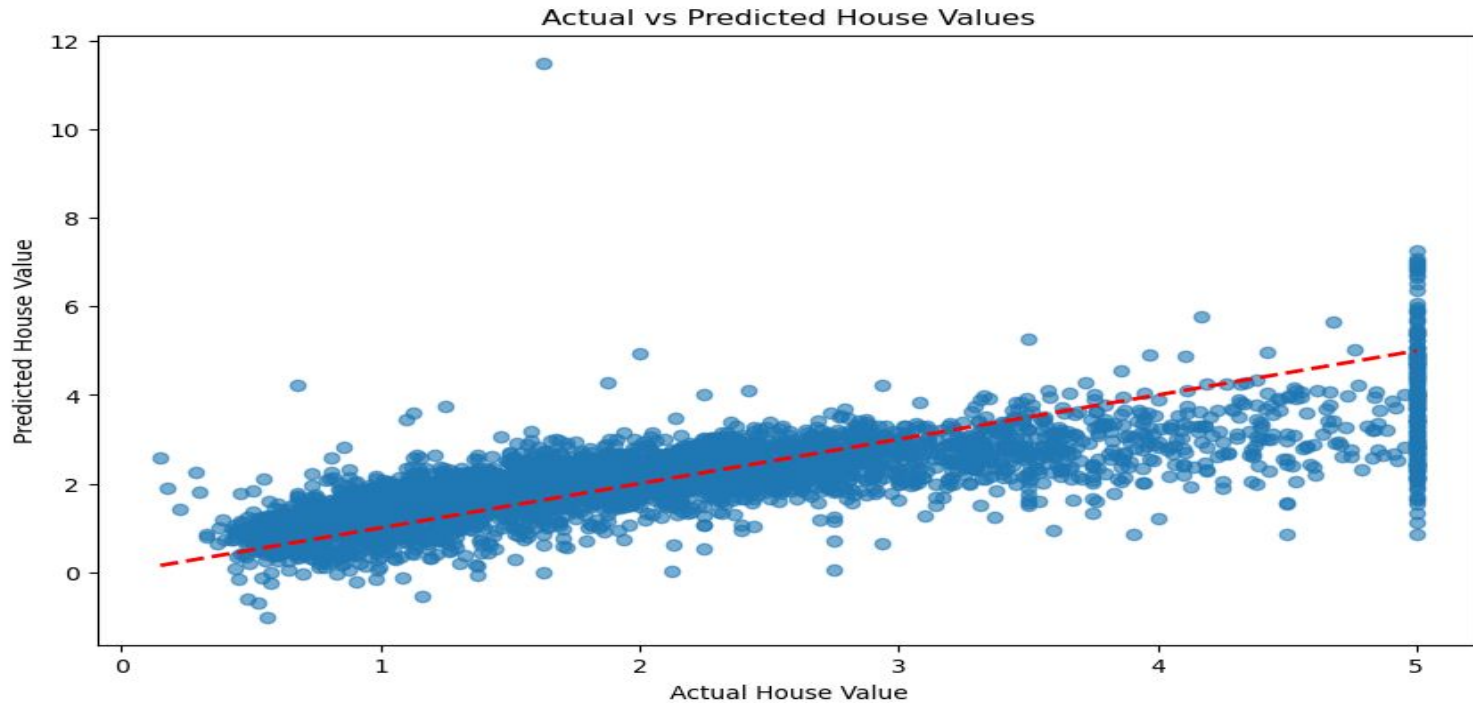
Mean Absolute Error (MAE): 0.5332

Root Mean Square Error (RMSE): 0.7456

$R^2$  Score: 0.5758

# Visualization

## Actual VS Predicted



# CONCLUSION

Successfully built and evaluated Linear Regression model.

Model explains ~57.6% of variance ( $R^2 = 0.5758$ ).

Good baseline performance.

Future Work: Try Random Forest, feature engineering, etc.